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=====
                        ITM TRACE SETUP
                        Real-Time Debug Output via SWO
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OVERVIEW
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=====
ITM (Instrumentation Trace Macrocell) enables real-time debug output via
SWO (Serial Wire Output) on ST-Link. Firmware writes to ITM stimulus ports,
OpenOCD captures SWO packets, nucleus trace decodes them to the dashboard.

```

```
Flow: Firmware (ITM) --SWO--> ST-Link --USB--> OpenOCD --TCP--> nucleus trace
```

```
=====
STEP 1: CONFIGURE TRACE IN stm32.toml
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```

```
File: stm32.toml
```

```
#####
[device]
family = "STM32F411RE"
board = "NUCLEO-F411RE"
clock_hz = 100_000_000
```

```
[build]
toolchain = "arm-none-eabi-gcc"
optimization = "Os"
```

```
[peripherals.usart2]
tx = "PA2"
rx = "PA3"
baud = 115200
```

```
[trace]
enabled = true
sw0_freq = 2_000_000
```

```
[[trace.variables]]
name = "blink_count"
port = 1
type = "u32"
#####
```

```
=====
STEP 2: FIRMWARE - ITM INITIALIZATION
=====
```

```
File: src/main.c (ITM Setup Code)
```

```
#####
#include "stm32f4xx_hal.h"
#include "generated/nucleus_config.h"

/* Initialize ITM for SWO trace output */
void itm_init(uint32_t core_hz, uint32_t sw0_hz)
{
    CoreDebug->DEMCR |= CoreDebug_DEMCR_TRCENA_Msk; /* enable tracing */
    TPI->SPPR = 2; /* SWO protocol */
    TPI->ACPR = (core_hz / sw0_hz) - 1; /* baud divisor */
    ITM->LAR = 0xC5ACCE55; /* unlock ITM */
    ITM->TCR |= ITM_TCR_ITMENA_Msk; /* enable ITM */
    ITM->TER |= (1UL << 0) | (1UL << 1); /* enable ports 0 (log) & 1 (vars) */
}

```

```
/* Log text via ITM port 0 (shows in dashboard) */
void itm_log(const char *s)
{
    while (*s) {
        while (ITM->PORT[0].u32 == 0) { } /* wait for FIFO space */
        ITM->PORT[0].u8 = (uint8_t)*s;
        s++;
    }
}

```

```
/* Write u32 to ITM port for variable tracing */
static inline void itm_write32(uint8_t port, uint32_t value)
{
    if (!(ITM->TCR & ITM_TCR_ITMENA_Msk) || !(ITM->TER & (1UL << port)))
        return;
    while (ITM->PORT[port].u32 == 0) { }
    ITM->PORT[port].u32 = value;
}

```

```
int main(void)
{
    HAL_Init();
    SystemClock_Config();
    Nucleus_Init();

    /* Initialize ITM: 100 MHz core clock, 2 MHz SWO rate */
    itm_init(100000000, 2000000);
    itm_log("System initialized\n");

    __HAL_RCC_GPIOA_CLK_ENABLE();

    GPIO_InitTypeDef led = {0};
    led.Pin = GPIO_PIN_5;
    led.Mode = GPIO_MODE_OUTPUT_PP;
    led.Pull = GPIO_NOPULL;
    led.Speed = GPIO_SPEED_FREQ_LOW;
    HAL_GPIO_Init(GPIOA, &led);

    itm_log("LED initialized on PA5\n");

    uint32_t blink_count = 0;
    while (1) {

```

```

    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    blink_count++;
    itm_write32(1, blink_count); /* trace blink_count on port 1 */
    itm_log("Blink #");
    itm_log_uint(blink_count); /* custom helper for numbers */
    itm_log("\n");
    HAL_Delay(500);
}
}

```

``` ===== STEP 3: VALIDATE & BUILD ===== ```

```

$ nucleus check
stm32.toml OK - 1 peripheral(s), no conflicts.

$ nucleus build
generated ./src/generated/nucleus_config.h
generated ./src/generated/nucleus_init.c
[ 4%] Building C object CMakeFiles/firmware.dir/src/main.c.obj
[ 12%] Linking C executable firmware
Generating firmware.bin / firmware.hex
[100%] Built target firmware

✓ Build successful

```

``` ===== STEP 4: CONNECT HARDWARE & START OPENOCD ===== ```

```

Connect NUCLEO-F411RE via ST-Link USB cable. Start OpenOCD:

$ openocd -f interface/stlink.cfg -f target/stm32f4x.cfg
Open On-Chip Debugger 0.12.0
...
Info : starting gdb server on 3333
Info : Listening on port 4444 for telnet connections

(Keep OpenOCD running in background / separate terminal)

```

``` ===== STEP 5: FLASH FIRMWARE ===== ```

```

$ nucleus flash
Flashed firmware.bin to NUCLEO-F411RE

LED will start blinking. ITM data flowing via SWO.

```

``` ===== STEP 6: RUN NUCLEUS TRACE TO READ DEBUG OUTPUT ===== ```

```

$ nucleus trace --trace-tcp 127.0.0.1:3344 \
                --openocd 127.0.0.1:4444 \
                --config stm32.toml

INFO: connecting to OpenOCD at 127.0.0.1:4444...
INFO: configuring OpenOCD (core_hz=100000000, swo_hz=2000000)
INFO: tpiu config internal :3344 uart off 100000000 2000000
INFO: itm ports on
INFO: listening for SWO packets on 127.0.0.1:3344
INFO: dashboard WebSocket listening on 127.0.0.1:5678
INFO: ready for browser connection

(Command runs and waits for browser connection)

```

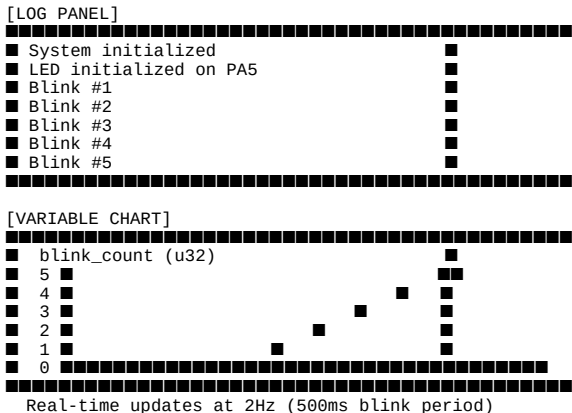
``` ===== STEP 7: OPEN TRACE DASHBOARD ===== ```

```

In VS Code:
- Run: Nucleus: Open Trace Dashboard
- Or manually: http://localhost:5678

```

Dashboard shows:



```

=====
RESULT: LIVE DEBUG OUTPUT ENABLED
=====

```

- ✓ ITM configured on firmware
- ✓ OpenOCD capturing SWO packets
- ✓ nucleus trace decoding to dashboard
- ✓ Log lines and variable traces visible in real-time

Dashboard persists as long as:

- nucleus trace is running
- OpenOCD has target connection
- Firmware continues running on board

Stop: Ctrl+C in nucleus trace, power off board, or disconnect ST-Link.

=====